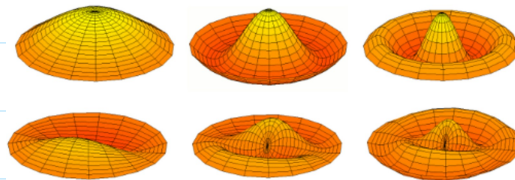


Two Dimensional Wave Equation

Saturday, July 26, 2025 5:09 PM

11. Two dimensional wave equation

11.1 A elastic membrane is basically just the skin on a drum (i.e drum head) that is stretched and fixed along its edge.



Drumhead Vibration Modes

<https://images.app.goo.gl/aPK9aARPKKnjdKfFA>

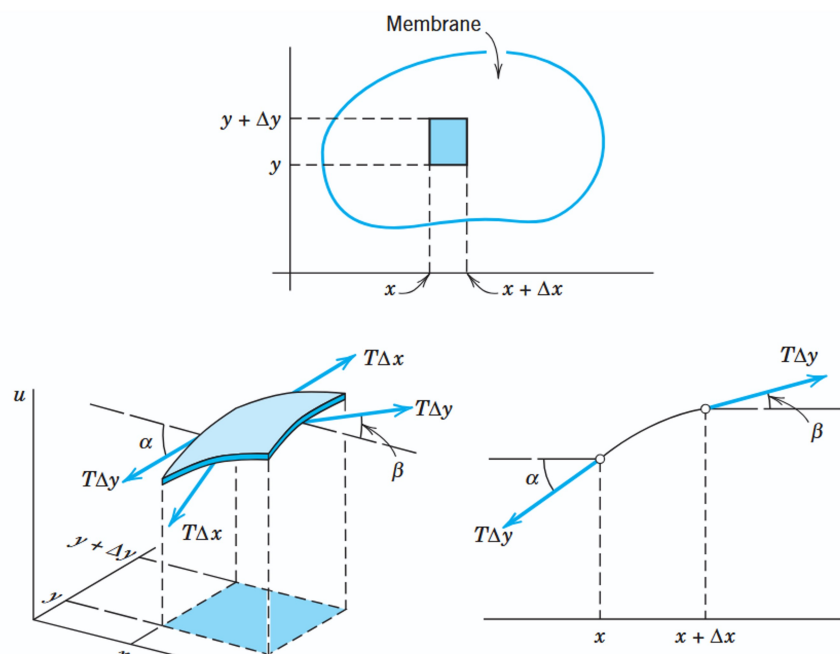


Physical Assumptions

1. The mass of the membrane per unit area is constant ("homogeneous membrane"). The membrane is perfectly flexible and offers no resistance to bending.
2. The membrane is stretched and then fixed along its entire boundary in the xy -plane. The tension per unit length T caused by stretching the membrane is the same at all points and in all directions and does not change during the motion.
3. The deflection $u(x, y, t)$ of the membrane during the motion is small compared to the size of the membrane, and all angles of inclination are small.

Although these assumptions cannot be realized exactly, they hold relatively accurately for small transverse vibrations of a thin elastic membrane, so that we shall obtain a good model, for instance, of a drumhead.

11.2 modeling:



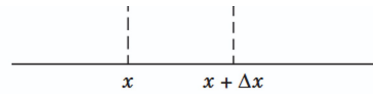
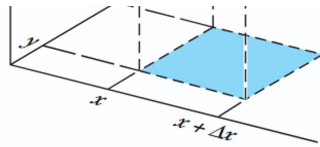


Fig. 301. Vibrating membrane

We are interested in the vertical components of tension T :

$$\begin{aligned} \text{for fixed } y: T\Delta y(\sin\beta - \sin\alpha) &\approx T\Delta y(\tan\beta - \tan\alpha) \\ &\approx T\Delta y[u_x(x + \Delta x, y) - u_x(x, y)] \end{aligned}$$

$$\text{for fixed } x: \approx T\Delta x[u_y(x, y + \Delta y) - u_y(x, y)]$$

Newton's 2nd law:

$$\underbrace{\rho \Delta x \Delta y}_{\text{mass}} \frac{\partial^2 u}{\partial t^2} = T\Delta y[u_x(x + \Delta x, y) - u_x(x, y)] + T\Delta x[u_y(x, y + \Delta y) - u_y(x, y)]$$

$$\Rightarrow \frac{\partial^2 u}{\partial t^2} = \frac{T}{\rho} \left[\frac{u_x(x + \Delta x, y) - u_x(x, y)}{\Delta x} + \frac{u_y(x, y + \Delta y) - u_y(x, y)}{\Delta y} \right]$$

As $\Delta x \rightarrow 0, \Delta y \rightarrow 0$, get

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$c^2 = \frac{T}{\rho}$$

12. Rectangular Membrane

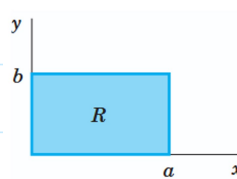
12.1 The Problem:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$u = 0 \text{ on the boundary}$$

$$u(x, y, 0) = f(x, y)$$

$$u_t(x, y, 0) = g(x, y).$$



we will set $g(x, y)$ to be zero, i.e., $u_t(x, y, 0) = 0$

12.2 Use separation of variables to obtain three ODEs

$$\text{Let } u(x, t) = F(x)G(y)H(t)$$

$$\text{Then } \frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \text{ becomes:}$$

Let $u(x,t) = F(x)G(y)H(t)$

Then $\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$ becomes:

$$F(x)G(y) \frac{d^2 H}{dt^2} = c^2 \left[G(y)H(t) \frac{d^2 F}{dx^2} + F(x)H(t) \frac{d^2 G}{dy^2} \right]$$

Divide by FGH to get:

$$\frac{1}{H} \frac{d^2 H}{dt^2} = c^2 \left[\frac{1}{F} \frac{d^2 F}{dx^2} + \frac{1}{G} \frac{d^2 G}{dy^2} \right]$$

First separation:

$$\left\{ \begin{array}{l} \text{Let } \frac{1}{H} \frac{d^2 H}{dt^2} = -k^2 c^2 = -\lambda^2 \quad (\text{i.e. } \lambda = kc) \\ \text{ie } \frac{d^2 H}{dt^2} + \lambda^2 H = 0 \end{array} \right.$$

and $\left\{ \begin{array}{l} \frac{1}{F} \frac{d^2 F}{dx^2} + \frac{1}{G} \frac{d^2 G}{dy^2} = -k^2 \\ \Rightarrow \frac{1}{F} \frac{d^2 F}{dx^2} = -\frac{1}{G} \frac{d^2 G}{dy^2} - k^2 = -\frac{1}{G} \left(\frac{d^2 G}{dy^2} + k^2 G \right) \end{array} \right.$

Second Separation:

$$\text{let } \left\{ \frac{1}{F} \frac{d^2 F}{dx^2} = -l^2 \Rightarrow \frac{d^2 F}{dx^2} + l^2 F = 0 \right.$$

$$\left\{ \begin{array}{l} -\frac{1}{G} \left(\frac{d^2 G}{dy^2} + k^2 G \right) = -l^2 \Rightarrow \frac{d^2 G}{dy^2} + k^2 G = l^2 G \\ \Rightarrow \frac{d^2 G}{dy^2} + (k^2 - l^2) G = 0 \end{array} \right.$$

12.3 Solving the DEs according to boundary conditions:

$$F(0) = F(a) = 0, \quad G(0) = G(b) = 0$$

gives us:

$$F(x) = B \sin\left(\frac{m\pi}{a} x\right) \quad (m \text{ integer}) \quad (\text{i.e. } l = m\pi/a)$$

$$F(x) = B \sin\left(\frac{m\pi}{a} x\right) \quad (m \text{ integer}) \quad (\text{i.e. } \ell = m\pi/a)$$

$$\text{and} \quad G(y) = C \sin\left(\frac{n\pi}{b} y\right) \quad (n \text{ integer}) \quad (\text{i.e. } k^2 - \ell^2 = \left(\frac{n\pi}{b}\right)^2)$$

$$\begin{aligned} \text{which leaves us with } u(x, y, t) &= F(x)G(y)H(t) \\ &= \underbrace{BC}_A \sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right) H(t) \end{aligned}$$

$$\text{Also: } \frac{d^2 H}{dt^2} + \lambda^2 H = 0 \Rightarrow H(t) = P \cos(\lambda t) + Q \sin(\lambda t)$$

$$\text{Now } \lambda = ck = c \sqrt{\left(\frac{n\pi}{b}\right)^2 + \left(\frac{m\pi}{a}\right)^2}, \text{ i.e. } \lambda_{mn} = c\pi \sqrt{\left(\frac{n}{b}\right)^2 + \left(\frac{m}{a}\right)^2}$$

Thus, we get a solution:

$$u_{m,n}(x, y, t) = A \sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right) \left[P \cos(\lambda_{mn} t) + Q \sin(\lambda_{mn} t) \right]$$

$$\Rightarrow u_{m,n}(x, y, t) = \left[A_{m,n} \cos\left(c\pi \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} t\right) + B_{m,n} \sin\left(c\pi \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} t\right) \right] \sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right)$$

$$\text{where } A_{m,n} = A \cdot P, \quad B_{m,n} = A \cdot Q$$

12.4 To match initial conditions: $u(x, y, 0) = f(x, y)$
and $u_t(x, y, 0) = 0$

we need the **Double Fourier Series**:

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} u_{m,n}(x, y, t)$$

$$= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \left[A_{m,n} \cos(\lambda_{mn} t) + B_{m,n} \sin(\lambda_{mn} t) \right] \sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right)$$

and get $B_{mn} = 0$ from $u_t(x, y, 0) = 0$

and

$$A_{mn} = \left(\frac{2}{b}\right) \int_0^b \left[\sin\left(\frac{n\pi y}{b}\right) \cdot \frac{2}{a} \int_0^a f(x, y) \sin\left(\frac{m\pi x}{a}\right) dx \right] dy$$

$$A_{mn} = \left(\frac{2}{b}\right) \int_0^b \left[\sin\left(\frac{n\pi y}{b}\right) \cdot \frac{2}{a} \int_0^a f(x,y) \sin\left(\frac{m\pi x}{a}\right) dx \right] dy$$

$y = \text{const}$

In other words:

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \cos(\lambda_{mn} t) \sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right)$$

with

$$A_{mn} = \frac{4}{ab} \int_0^b \int_0^a f(x,y) \sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right) dx dy$$

125 Example:

Find the deflection $u(x, y, t)$ of the square membrane of side π and $c^2 = 1$ for initial velocity 0 and initial deflection

$$f(x, y) = 0.1 \sin 2x \sin 4y$$

Soln: Here $a = b = \pi$

We know: $A_{mn} = \frac{4}{ab} \int_0^b \int_0^a f(x, y) \sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right) dx dy$

$$= \frac{4}{\pi^2} \int_0^{\pi} \int_0^{\pi} (0.1 \sin(2x) \sin(4y)) \sin(mx) \sin(ny) dx dy$$

Since $\frac{2}{\pi} \int_0^{\pi} \sin(2x) \sin(mx) dx = \begin{cases} 1 & \text{if } m=2 \\ 0 & \text{otherwise} \end{cases}$ and $\frac{2}{\pi} \int_0^{\pi} \sin(4y) \sin(ny) dy = \begin{cases} 1 & \text{if } n=4 \\ 0 & \text{otherwise} \end{cases}$

Thus

$$A_{2,4} = 0.1 \text{ and all other } A_{m,n} = 0$$

In that case, for $m=2, n=4, c=1$

$$\lambda_{2,4} = c\pi \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = (1)\pi \sqrt{\left(\frac{2}{\pi}\right)^2 + \left(\frac{4}{\pi}\right)^2} = (1) \sqrt{4+16} = \sqrt{20} = 2\sqrt{5}$$

Therefore:

$$u(x, y, t) = 0.1 \cos(2\sqrt{5} t) \sin(2x) \sin(4y)$$